PRINCIPAL COMPONENT ANALYSIS (PCA)

Unsupervised Machine Learning Models

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2026-10-13

SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(ggthemes)
library(psych)
library(sjmisc)
library(kableExtra)
library(knitr)
library(tidyr)
library(reshape2)
library(formatR)
library(corrplot)
library(e1071)
library(caret)
library(Metrics)
library(xgboost)
library(readxl)
```

Explanation

No Target Variable(There is no Prediction)

- goal: the main goal of unsupervised machine learning is to reduce dimension(dimension reduction using pca) and classify the variables into groups or clusters using k-means clustering

load the dataset

```
data <- read_excel("PCA data.xlsx")
head(data)
```

```
# A tibble: 6 × 10
  Study_Hours Attendance Assignment_Score Quiz_Score Exam_Score Readin
g_Hours
        <dbl>      <dbl>            <dbl>      <dbl>      <dbl>
<dbl>
1        1.21       71.7             41.9       33.8       62.0
0.8
2        2.6        66.0             63.2       62.0       50.5
2.14
3        4.21       83.8             81.0       81.5       80.7
2.21
4        3.47       74.6             72.6       75.1       62.6
2.35
5        3.87       82.6             82.6       68.6       74.5
1.18
6        4.39       82.1             75.9       62.6       65.5
1.77
# i 4 more variables: Sleep_Hours <dbl>, Motivation <dbl>,
#   Internet_Usage <dbl>, Stress_Level <dbl>
```

confirm names

```
names(data)
```

```
 [1] "Study_Hours"      "Attendance"       "Assignment_Score" "Quiz_Sc
ore"
 [5] "Exam_Score"       "Reading_Hours"    "Sleep_Hours"      "Motivat
ion"
 [9] "Internet_Usage"   "Stress_Level"
```

check data structure

```
str(data)
```

```
tibble [100 × 10] (S3: tbl_df/tbl/data.frame)
 $ Study_Hours     : num [1:100] 1.21 2.6 4.21 3.47 3.87 4.39 1.52 3.8
9 2.87 3.66 ...
 $ Attendance      : num [1:100] 71.7 66 83.8 74.5 82.6 ...
 $ Assignment_Score: num [1:100] 41.9 63.2 81 72.6 82.6 ...
 $ Quiz_Score      : num [1:100] 33.8 62 81.5 75.1 68.5 ...
 $ Exam_Score      : num [1:100] 62 50.5 80.7 62.6 74.5 ...
 $ Reading_Hours   : num [1:100] 0.8 2.14 2.21 2.35 1.18 1.77 1.75 2.1
9 2.08 1.13 ...
 $ Sleep_Hours     : num [1:100] 6.52 8.6 7.45 7.51 7.72 7.44 6.47 7.8
6 8.2 8.95 ...
 $ Motivation      : num [1:100] 5.64 4.84 7.56 6.19 7.97 5.8 4.71 8.0
5 7.19 8.62 ...
 $ Internet_Usage  : num [1:100] 2.62 2.43 1.74 2.08 4.37 4.46 3.02 2.
72 0.36 4.76 ...
 $ Stress_Level    : num [1:100] 5.63 2.81 7.54 6.05 4.5 2.73 5.02 2.5
6 4.82 4.05 ...
```

check number of rows and columns

```
dim(data)
```

```
[1] 100  10
```

```
cat("NO of rows:", nrow(data), "\n", "No of columns:", ncol(data))
```

```
NO of rows: 100
 No of columns: 10
```

summary statistics

```
summary(data)
```

```
 Study_Hours      Attendance    Assignment_Score   Quiz_Score
Min.   :0.500   Min.   :48.43   Min.   :35.93    Min.   :33.13
1st Qu.:2.337   1st Qu.:67.65   1st Qu.:60.47    1st Qu.:52.54
Median :2.960   Median :74.72   Median :66.79    Median :60.09
Mean   :3.100   Mean   :74.99   Mean   :66.02    Mean   :60.50
3rd Qu.:3.890   3rd Qu.:82.72   3rd Qu.:72.38    3rd Qu.:68.39
Max.   :5.930   Max.   :99.77   Max.   :91.27    Max.   :86.95
  Exam_Score    Reading_Hours    Sleep_Hours      Motivation
Min.   :28.04   Min.   :0.100   Min.   :4.720   Min.   :1.020
1st Qu.:55.06   1st Qu.:1.157   1st Qu.:6.518   1st Qu.:4.940
Median :63.18   Median :1.690   Median :7.165   Median :6.285
Mean   :62.97   Mean   :1.631   Mean   :7.087   Mean   :6.309
3rd Qu.:71.27   3rd Qu.:2.112   3rd Qu.:7.620   3rd Qu.:7.508
Max.   :85.77   Max.   :3.050   Max.   :8.950   Max.   :9.750
Internet_Usage   Stress_Level
Min.   :0.200   Min.   :2.010
1st Qu.:2.025   1st Qu.:4.058
Median :2.700   Median :5.050
Mean   :2.812   Mean   :5.000
3rd Qu.:3.670   3rd Qu.:5.970
Max.   :6.280   Max.   :8.080
```

check missing values and dupliccates

check missing values

```
sum(is.na(data))
```

```
[1] 0
```

```
colSums((is.na(data)))
```

```
     Study_Hours       Attendance Assignment_Score       Quiz_Score
               0                0                0                0
      Exam_Score    Reading_Hours      Sleep_Hours       Motivation
               0                0                0                0
  Internet_Usage     Stress_Level
               0                0
```

check for duplicated rows

```
sum(duplicated(data))
```

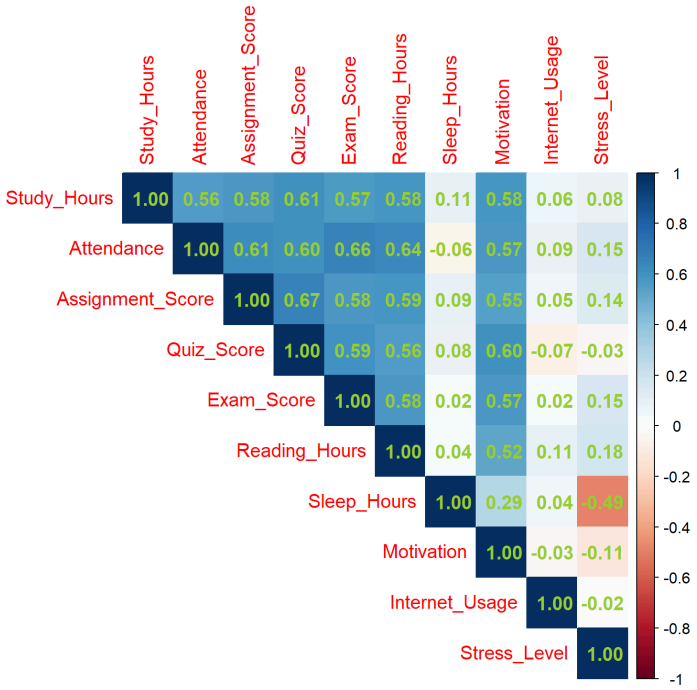
```
[1] 0
```

examine the correlation

```
correlation <- cor(data)
round(correlation, 2)
```

```
                Study_Hours Attendance Assignment_Score Quiz_Score Ex
am_Score
Study_Hours            1.00       0.56             0.58       0.61
0.57
Attendance             0.56       1.00             0.61       0.60
0.66
Assignment_Score       0.58       0.61             1.00       0.67
0.58
Quiz_Score             0.61       0.60             0.67       1.00
0.59
Exam_Score             0.57       0.66             0.58       0.59
1.00
Reading_Hours          0.58       0.64             0.59       0.56
0.58
Sleep_Hours            0.11      -0.06             0.09       0.08
0.02
Motivation             0.58       0.57             0.55       0.60
0.57
Internet_Usage         0.06       0.09             0.05      -0.07
0.02
Stress_Level           0.08       0.15             0.14      -0.03
0.15
                Reading_Hours Sleep_Hours Motivation Internet_Usage
Study_Hours              0.58        0.11       0.58           0.06
Attendance               0.64       -0.06       0.57           0.09
Assignment_Score         0.59        0.09       0.55           0.05
Quiz_Score               0.56        0.08       0.60          -0.07
Exam_Score               0.58        0.02       0.57           0.02
Reading_Hours            1.00        0.04       0.52           0.11
Sleep_Hours              0.04        1.00       0.29           0.04
Motivation               0.52        0.29       1.00          -0.03
Internet_Usage           0.11        0.04      -0.03           1.00
Stress_Level             0.18       -0.49      -0.11          -0.02
                Stress_Level
Study_Hours             0.08
Attendance              0.15
Assignment_Score        0.14
Quiz_Score             -0.03
Exam_Score              0.15
Reading_Hours           0.18
Sleep_Hours            -0.49
Motivation             -0.11
Internet_Usage         -0.02
Stress_Level            1.00
```

```
corrplot(correlation, method = "color", type = "upper", addCoef.col =
"yellowgreen")
```



why check correlation?

- purpose: to seee which variables are related,. PCA is particulalry useful when variables share overlapping information.

standardize the variable

```
data_scaled <- scale(data)
cat("The mean of standardized data.:\n")
```

```
The mean of standardized data.:
```

```
round(colMeans(data_scaled), 2)
```

```
     Study_Hours       Attendance Assignment_Score       Quiz_Score 
               0                0                0                0 
      Exam_Score   Reading_Hours      Sleep_Hours       Motivation 
               0                0                0                0 
  Internet_Usage     Stress_Level 
               0                0 
```

```
cat("The sd of standardized data.:\n")
```

```
The sd of standardized data.:
```

```
round(apply(data_scaled, 2, sd, 2))
```

```
     Study_Hours       Attendance Assignment_Score       Quiz_Score 
               1                1                1                1 
      Exam_Score    Reading_Hours      Sleep_Hours       Motivation 
               1                1                1                1 
  Internet_Usage     Stress_Level 
               1                1 
```

perform pca

```
pca_results <- prcomp(data, center = TRUE, scale. = TRUE, )
summary(pca_results)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4     PC5     PC6
PC7
Standard deviation     2.1351 1.2659 1.0201 0.74406 0.68761 0.66518 0.
64206
Proportion of Variance 0.4559 0.1603 0.1041 0.05536 0.04728 0.04425 0.
04122
Cumulative Proportion  0.4559 0.6161 0.7202 0.77555 0.82283 0.86708 0.
90830
                           PC8     PC9    PC10
Standard deviation     0.59486 0.54569 0.51510
Proportion of Variance 0.03539 0.02978 0.02653
Cumulative Proportion  0.94369 0.97347 1.00000
```

component loading

```
pca_results$rotation
```

```
                       PC1          PC2          PC3         PC4
PC5
Study_Hours      0.37159443  0.024278333 -0.028740019  0.17328351  0.2
39144275
Attendance       0.38555408 -0.126843702 -0.065731741 -0.26758306 -0.3
21721430
Assignment_Score 0.38150181 -0.026930674 -0.008500817  0.10341776  0.5
48714137
Quiz_Score       0.38337408  0.072997272  0.174488939 -0.31996465  0.4
31252630
Exam_Score       0.37853350 -0.078994216  0.023843107 -0.05605759 -0.4
85418432
Reading_Hours    0.37202203 -0.095013109 -0.123698342  0.18729395 -0.1
07454304
Sleep_Hours      0.05257602  0.671011346 -0.069731260  0.60249987 -0.0
57528816
Motivation       0.36460328  0.239652828  0.100370936  0.03443770 -0.3
13047444
Internet_Usage   0.02458860  0.005455497 -0.966185664 -0.10860350  0.0
61230066
Stress_Level     0.05247790 -0.674003213  0.003662049  0.60917262 -0.0
02105259
                        PC6         PC7         PC8         PC9
PC10
Study_Hours       0.799008408 -0.10678843 -0.16117022  0.29501813 -0.1
0225092
Attendance       -0.209877444  0.12595661  0.16910480  0.71972690  0.2
2207170
Assignment_Score -0.468987729 -0.14461386  0.09782126  0.10291788 -0.5
2995534
Quiz_Score       -0.068591019 -0.05006274 -0.09727390 -0.29266251  0.6
5312216
Exam_Score       -0.125402081 -0.42687446 -0.57879256 -0.22849370 -0.1
6116022
Reading_Hours    -0.007763996  0.82325463 -0.14076685 -0.28819828 -0.0
9981662
Sleep_Hours      -0.221877601 -0.03419124 -0.14133790  0.17141449  0.2
7575967
Motivation        0.150157162 -0.17503505  0.72393216 -0.34009937 -0.0
8626811
Internet_Usage    0.011488854 -0.14553314  0.06458508 -0.12811147  0.0
9171362
Stress_Level     -0.072027815 -0.19067123  0.15282333 -0.04453512  0.3
2399807
```

```
head(pca_results$x)
```

```
            PC1        PC2          PC3         PC4        PC5
PC6
[1,] -3.1004403 -0.8324155 -0.009257163 -0.02667295 -2.1907451  0.0946
0053
[2,] -0.9476065  2.1644286  0.056857635  0.36950154  0.6569267 -0.3334
2871
[3,]  3.1391220 -0.9734800  1.058853592  1.09172137  0.3862430 -0.5307
8563
[4,]  1.2304389 -0.2166684  0.617645514  0.75118338  0.8243416 -0.3054
9019
[5,]  1.8905492  0.9082887 -1.039322451 -0.19893562  0.4142862 -0.4993
2921
[6,]  1.0808669  1.2215125 -1.444495791 -0.83746544  0.6670605  0.2479
2530
             PC7        PC8         PC9       PC10
[1,] -0.38775060  0.2972036  0.16964191 -0.1013859
[2,]  1.49333406 -0.7497907 -0.03501788  0.1949122
[3,] -0.58869532 -0.3400872 -0.28992413  0.7704904
[4,]  0.64766381 -0.2939158 -0.39568609  0.6937774
[5,] -1.52804958  0.2654593  0.25133922 -0.2534581
[6,]  0.06228619 -0.5848292  0.78570420 -0.6665863
```

PURPOSE:

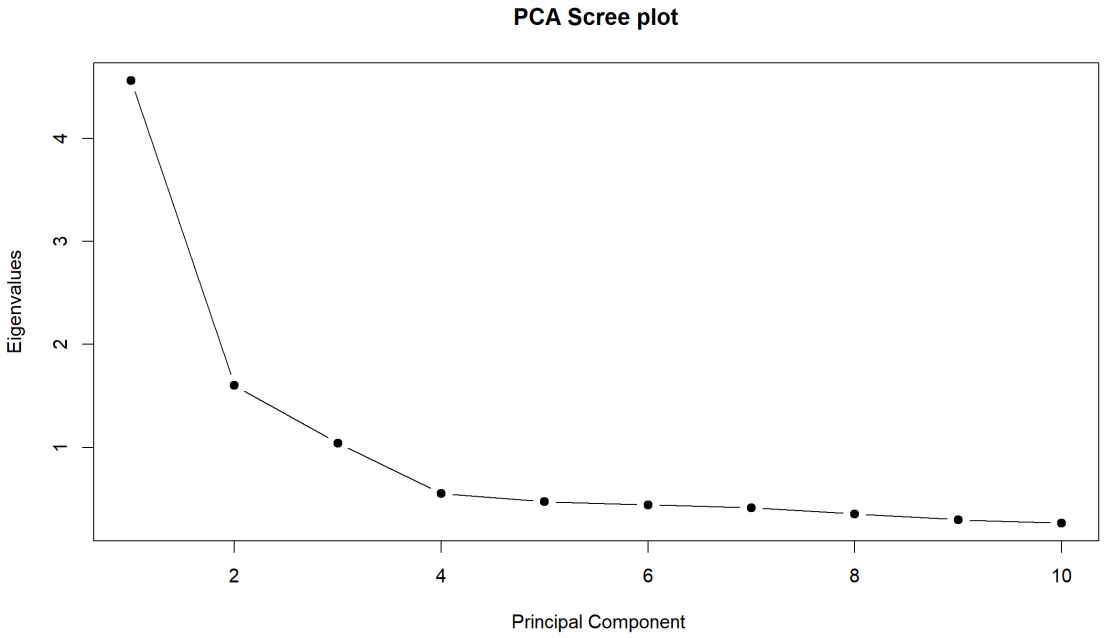
- THIS CREATES NEW VARIABLES

```
eigenvalues <- pca_results$sdev^2
variance <- eigenvalues/sum(eigenvalues) * 100
cumulative <- cumsum(variance)
results <- data.frame(component = paste0("pc", seq_along(eigenvalue
s)), Eigenvalue = round(eigenvalues,
    3), Variance_percentage = round(variance, 2), Cumulative_percent =
round(cumulative,
    2))
print(results)
```

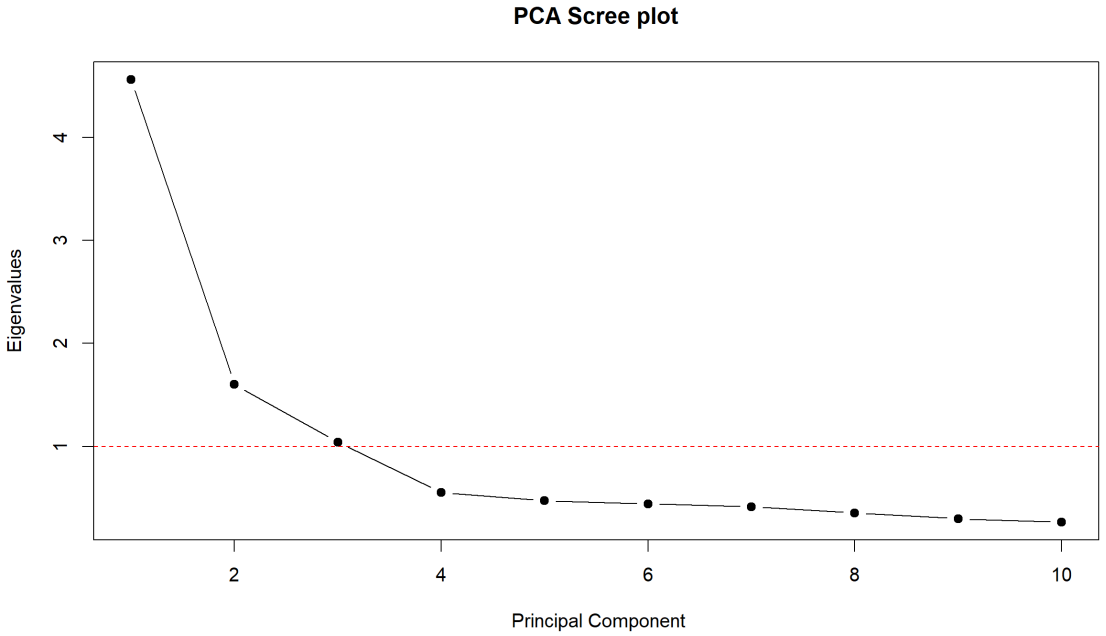
```
   component Eigenvalue Variance_percentage Cumulative_percent
1        pc1      4.559               45.59              45.59
2        pc2      1.603               16.03              61.61
3        pc3      1.041               10.41              72.02
4        pc4      0.554                5.54              77.56
5        pc5      0.473                4.73              82.28
6        pc6      0.442                4.42              86.71
7        pc7      0.412                4.12              90.83
8        pc8      0.354                3.54              94.37
9        pc9      0.298                2.98              97.35
10      pc10      0.265                2.65             100.00
```

draw the scree plot

```
plot(eigenvalues, type = "b", pch = 19, xlab = " Principal Component",
ylab = "Eigenvalues",
    main = "PCA Scree plot")
```




```
abline(h = 1, lty = 2, col = "red", lwd = 1)
```



select number of components

```
number_compopnents <- which(cumulative >= 80)[1]
cat("components retained:", number_compopnents, "\n")
```

```
components retained: 5
```

```
cat("cummulative variance explained:", round(cumulative[number_compopn
ents], 2),
    "%")
```

```
cummulative variance explained: 82.28 %
```

interplet component loadings

```
loadings <- pca_results$rotation[, 1:number_compopnents, drop = FALSE]
round(loadings * 100, 3)
```

```
                    PC1     PC2     PC3     PC4     PC5
Study_Hours      37.159   2.428  -2.874  17.328  23.914
Attendance       38.555 -12.684  -6.573 -26.758 -32.172
Assignment_Score 38.150  -2.693  -0.850  10.342  54.871
Quiz_Score       38.337   7.300  17.449 -31.996  43.125
Exam_Score       37.853  -7.899   2.384  -5.606 -48.542
Reading_Hours    37.202  -9.501 -12.370  18.729 -10.745
Sleep_Hours       5.258  67.101  -6.973  60.250  -5.753
Motivation       36.460  23.965  10.037   3.444 -31.305
Internet_Usage    2.459   0.546 -96.619 -10.860   6.123
Stress_Level      5.248 -67.400   0.366  60.917  -0.211
```

Attain the main goal of pca

reduce dataset from 10 original variables to the number of components selected

extract the retained compobnents

```
reduced_data <- as.data.frame(pca_results$x[, 1:number_compopnents, dr
op = FALSE])
head(reduced_data)
```

```
         PC1        PC2          PC3         PC4        PC5
1 -3.1004403 -0.8324155 -0.009257163 -0.02667295 -2.1907451
2 -0.9476065  2.1644286  0.056857635  0.36950154  0.6569267
3  3.1391220 -0.9734800  1.058853592  1.09172137  0.3862430
4  1.2304389 -0.2166684  0.617645514  0.75118338  0.8243416
5  1.8905492  0.9082887 -1.039322451 -0.19893562  0.4142862
6  1.0808669  1.2215125 -1.444495791 -0.83746544  0.6670605
```

check new dimension of data set]

```
dim(reduced_data)
```

```
[1] 100   5
```

save the reduced data set

```
library(writexl)
write.csv(reduced_data, "PCA_Reduced_Datase.xlsx", row.names = FALSE)
```